# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

## CLASS 6 — MATHS UNIT TEST SET A

Chapters 1–5: **Patterns in Mathematics, Lines and Angles, Number Play, Data Handling and Presentation, Prime Time** | RBSE Pattern | Session 2026–27

**Time: 2 Hours** Name: _____ Roll No.: _____ **Maximum Marks: 50**

**General Instructions:**

1. All questions are compulsory.
2. Q. 1–12 are Objective Type (1 mark each).
3. Q. 13–17 are Very Short Answer Type (2 marks each).
4. Q. 18–23 are Short Answer Type (3 marks each).
5. Q. 24–25 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 22 to 25.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

## SECTION — A (Objective Type Questions) 12 × 1 = 12

*Choose the correct option and write it in your answer book.*

**1.** Which of these is NOT a square number? 1

**(a)** 64 **(b)** 49 **(c)** 50 **(d)** 81

**2.** The 8th square number is: 1

**(a)** 81 **(b)** 72 **(c)** 64 **(d)** 56

**3.** The next number in the triangular number sequence $1, 3, 6, 10, 15, \ldots$ is: 1

**(a)** 24 **(b)** 20 **(c)** 22 **(d)** 21

**4.** A line segment has: 1

**(a)** two end points **(b)** one end point **(c)** no end point **(d)** three end points

**5.** The angle between the hands of a clock at 3 o'clock is: 1

**(a)** 180° **(b)** 90° **(c)** 60° **(d)** 120°

**6.** An angle of 89° is: 1

**(a)** acute **(b)** reflex **(c)** obtuse **(d)** right

**7.** The Kaprekar constant for 4-digit numbers is: 1

**(a)** 1729 **(b)** 9999 **(c)** 6174 **(d)** 1089

**8.** The Collatz sequence starting at 10 takes how many steps to reach 1? 1

**(a)** 5 **(b)** 9 **(c)** 6 **(d)** 7

**9.** Information collected for a purpose, such as marks or heights, is called: 1

**(a)** a graph **(b)** a scale **(c)** an angle **(d)** data

**10.** Which of the following is a good habit when drawing graphs? 1

**(a)** Label both axes and write a title **(b)** Skip the scale

**(c)** Leave bars unlabelled **(d)** Use different gaps

**11.** The HCF is the: 1

**(a)** half of the common factor **(b)** highest common fraction

**(c)** highest common multiple **(d)** highest common factor

**12.** The HCF of 10 and 35 is: 1

**(a)** 5 **(b)** 6 **(c)** 10 **(d)** 4

## SECTION — B (Very Short Answer Type) — 5 × 2 = 10

*Answer in about 10–15 words each.*

**13.** Write the first five triangular numbers. 2

**14.** How many degrees does the hour hand turn in 2 hours? 2

**15.** Is the sum of two odd numbers odd or even? Explain with an example. 2

**16.** In a pictograph, a symbol stands for 20. How many does 4½ symbols represent? 2

**17.** Define a prime number. Give three examples. 2

## SECTION — C (Short Answer Type) — 6 × 3 = 18

*Answer in about 30–40 words each.*

**18.** Write the rule and the next three terms of 4, 7, 10, 13, … 3

**19.** Explain with an example how a palindrome is formed by adding a number to its reverse repeatedly. 3

**20.** A bar graph shows rainfall: Mon 20 mm, Tue 35 mm, Wed 10 mm. Find the total and the difference between the highest and lowest. 3

**21.** Express 90 as a product of primes. How many prime factors does it have? 3

**22.** What is wrong with a bar graph whose bars have different widths and uneven gaps? 3

OR

Which fruit is liked by most children if the counts are Apple 12, Banana 9, Mango 15, Orange 6? Find the total. 3

**23.** State and test the divisibility rule of 9 on 7,128. 3

OR

Explain with an example why two consecutive numbers are always co-prime. 3

## SECTION — D (Long Answer Type) — 2 × 5 = 10

*Answer in about 80–100 words each.*

**24.** (i) List the types of angles with their measures.
(ii) Give one example of each from daily life. 5

OR

(i) Explain parallel and perpendicular lines with diagrams.
(ii) Give two examples of each. 5

**25.** (i) State the divisibility tests for 2, 3, 4, 5, 6, 9, 10.
(ii) Test 5,940. 5

OR

(i) Find the prime factorisation of 180.
(ii) Find its number of factors.
(iii) List all its factors. 5

## END OF QUESTION PAPER — Total: 50 Marks